

Maximilian Slavik

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Engineering Portfolio - slavikm.dev

Rensselaer Polytechnic Institute - Aerospace and Mechanical Engineering '27

Epistemically curious and driven student, leading complex engineering projects and developing novel mathematical methods for astrodynamics. Dean's List (Spring and Fall 2024). Four-year scholar-athlete in high school. Eagle Scout.

Experience:

➤ **Propulsion Intern - Orbital Operations Inc.** (May 2026 to Present):

- Built in-house code modeling the complex supercritical flow thermodynamics within the engine thrust chamber assembly. Developed numerical techniques specialized to pseudocritical transitions and refined my code's data structure, improving the code's runtime by a factor of seven and increasing accuracy simultaneously.
- Wrote a separate codebase that uses known physics to quickly turn test data into accurate numerical models, allowing the entire regenerative cooling model to be re-tuned in minutes.
- *Accepted an offer to stay onboard and continue working in the fall virtually.*

➤ **Mechanical Engineering Intern, Polymathic Innovations** (May 2025 to August 2025):

- Responsible for the comprehensive aerodynamic and mechanical design of a modular wind turbine unit. Field tests showed that our innovative approach to small wind nearly *doubled turbine efficiency* in certain cases, while maintaining a low-cost accessible product.

➤ **Chief Engineer, RPI Experimental Propulsion Initiative** (August 2024 to Present):

- Leading a team of over 60 engineers in the technical design and testing of an advanced regeneratively cooled liquid rocket engine and its flight vehicle. First liquid bipropellant rocket engine at RPI (previous project, currently in testing phase).

➤ **Subsystems Lead, RPI Design Build Fly** (August 2024 to May 2025):

- Design and manufacturing of subsystem components for an RC plane built for the AIAA Design Build Fly competition. Our team placed 5th in the global competition, with the second highest mission score.

➤ **Mechanical Design Lead, Undergraduate Jet Engine Project** (August 2025 to Present):

- Design and simulation of a power transfer assembly for turbomachinery and an annular combustion chamber structure. High-frequency modal analysis and thermal management. Once it is tested by our undergraduate team, this engine will be used in an RPI propulsion systems lab course.

➤ **Undergraduate Research: Symplectic Jacobian Reconstruction Methods** (August 2025 to Present):

- Development of an efficient numerical algorithm based on group theory that analyzes data from a Hamiltonian dynamical system and generates a best-fit locally linear system that describes it.
- Key Skills: MATLAB, Applied mathematics (linear algebra, differential equations, group theory, dynamical systems, orthogonal functions), LaTeX document formatting software.

Projects:

- **Stardust - Personal Regeneratively Cooled Liquid Rocket** (Summer 2026):
 - Designed a 3D-printed, regeneratively cooled liquid rocket engine, high-pressure propellant tanks, and main valves for a personal flight vehicle. Went from idea to complete CAD and analysis in under two and a half months.
 - Fabricating static-fire hardware in the RPI machine shop; igniter testing is complete. Targeting a flight to a projected 24,000 ft AGL in Spring 2027.
- **Cislunar Invariant Manifolds in the Circular Restricted Three Body Problem** (Spring 2026):
 - Wrote a Python tool that computes the stable and unstable invariant manifolds branching off Earth-Moon Halo, Dragonfly, and Axial orbits. Orbit starting conditions came from JPL's periodic orbit database.
- **Pneumatic Spool Valve** [Engineering Design Course Project, Spring 2026]:
 - Worked in a team of seven students to design, machine, and test an automated alternative to conventional industrial globe valves. The valve was able to achieve a similar flow coefficient to several industrial globe valves on the market, while being fully automated, mechanically fail-safed, and substantially cheaper.

Skills:

- **CAE:** Python, Siemens NX CAD, Siemens NX drafting, Simcenter NASTRAN finite element modeling (Structural, Thermal, Vibro-Acoustic, Fluids, Space Systems), MATLAB, Simulink, Algorithmic Modeling, Rocket Propulsion Analysis (RPA)
- **Applied Mathematics:** Optimal Control theory, partial differential equations, Fourier analysis, calculus of variations, dynamical systems, Hamiltonian and Lagrangian mechanics, linear algebra, group theory, statistics, quantum mechanics, field theory (classical and quantum).
- **Manufacturing:** Milling (CNC and manual), 3D printing, lathe (CNC and manual), dimensioning.
- **Leadership:**
 - Was awarded the Eagle Scout rank, the highest achievable rank in Scouting. This rank demonstrates well-refined leadership, time management, and communication.
 - Earned Brotherhood rank in the Order of the Arrow, a peer-elected national honor society in scouting.
 - Served as team captain of my high school's varsity swim team during the Fall of 2023

Education and Relevant Coursework

B.S.E. Mechanical and Aerospace Engineering, Dean's Honor List (S24, F24), Cumulative 3.25 GPA

- **Engineering:** Spaceflight Mechanics, Aerodynamics, Propulsion Systems, Aeroelasticity, Modeling and Control of Dynamical Systems, Heat Transfer, Aerospace Structures, Elements of Mechanical Design, Fluid Mechanics, Numerical Methods and Programming, Strengths of Materials (structural deformation and stress), Thermodynamics, Engineering Dynamics, Statics, Mechatronics, Engineering Graphics, Materials Science.
- **Physics and Mathematics:** Calculus (I, II, Multivariable), Matrix Algebra, Modeling and Analysis of Uncertainty, Differential Equations, Physics I, Physics II (Electromagnetism and Relativity).
- **Non-Technical:** Professional Development III: Leadership Competencies, Professional Development II - Tech Issues and Solutions, Principles of Economics, Media and Society, Human Memory, Psychology

